

PRIYANKA DESHPANDE

AI / ML Engineer • Data Engineer • GenAI & LLM Specialist • Data Analytics

Las Vegas, NV (89052) | deshpande.priya07@gmail.com | +1 (702) 850-9941

github.com/Inlighted | [linkedin.com/in/priyade](https://www.linkedin.com/in/priyade) | portfolio-z8g8.vercel.app | <https://leetcode.com/u/Impriya29/>

Authorized to work in the USA (EAD holder) — no visa sponsorship required. Open to relocation.

PROFESSIONAL SUMMARY

- 5+ years of AI/ML and Data Engineering experience owning problems end-to-end from raw data ingestion through model deployment to executive-ready dashboards across classical machine learning, deep learning, NLP and modern Generative AI.
- Specializes in Large Language Models and Generative AI: designs and deploys RAG pipelines, vector-search systems and prompt-engineered solutions for knowledge assistants, document Q&A and intelligent automation.
- Reduces hallucinations and improves grounding in LLM applications through evaluation pipelines, retrieval strategies and practical guardrails, with experience in human-in-the-loop and tool-augmented LLM systems.
- Builds scalable data-engineering pipelines on Azure Databricks and PySpark, processing multi-terabyte datasets for downstream ML training and real-time inference.
- Skilled in PySpark, Python, and SQL, with hands-on work in Delta Lake, medallion architecture, identity resolution, and data governance with Unity Catalog.
- Comfortable bridging data engineering and machine learning: feature engineering, MLflow tracking, model deployment, and monitoring.
- Designs distributed ETL/ELT pipelines that keep features consistent between training and serving environments, eliminating training-serving skew in production.
- Strong classical-ML and statistical-modeling foundation across text, tabular and image data — classification, regression, clustering and time-series forecasting with rigorous evaluation (F1, AUC-ROC, RMSE, cross-validation).
- Applies MLOps practices including experiment tracking, CI/CD integration, model versioning, drift monitoring and Blue-Green deployments for stable production releases.
- Delivers secure, governed GenAI in regulated environments: LLM evaluation, hallucination tracking, retrieval-quality validation, fairness checks, explainability and audit-ready documentation.
- Translates AI initiatives into measurable business value through automation, cost reduction, risk mitigation and operational efficiency.
- Certified Scrum Master who drives Agile delivery and collaborates with product, engineering and business teams to define problems, align on metrics and ship impactful solutions.
- Currently works as an AI/ML and Data Engineering consultant and tutor, mentoring aspiring data professionals in Python, SQL, ML and interview preparation.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, PySpark
- **Machine Learning:** Supervised & Unsupervised Learning, XGBoost, LightGBM, CatBoost, Random Forest, SVM, K-Means, DBSCAN, PCA, Ensemble Methods, Anomaly Detection, Classification, Regression, Clustering, Time-Series Forecasting
- **Deep Learning:** TensorFlow, PyTorch, Keras, CNN, RNN, LSTM, Attention Mechanism, BERT, GPT
- **NLP:** Hugging Face Transformers, spaCy, NLTK, LangChain, LlamaIndex, Text Classification, NER, Summarization, Q&A Systems, Retrieval-Augmented Generation (RAG)
- **Generative AI & LLMs:** Prompt Engineering, Fine-tuning LLMs (LoRA/Adapters), Synthetic Data Generation, LangChain, LangGraph, Crew AI, OpenAI APIs, Google Gemini, LLaMA, Claude (Anthropic)
- **Vector Databases:** FAISS, Pinecone, Weaviate, ChromaDB, Elasticsearch, Azure Cognitive Search
- **Agents & Orchestration:** LangChain Agents, Crew AI, OpenAI Function Calling, LangGraph, Agentic Frameworks
- **MLOps & Deployment:** MLflow, DVC, Docker, FastAPI, Flask, Streamlit, Azure ML, Jenkins, CI/CD Pipelines, Model Serving, Blue-Green Deployments, A/B Testing, Inference Optimization, Observability
- **Data Engineering:** Azure Databricks, Apache Spark, PySpark, Apache Airflow, Azure Data Factory, Azure Synapse Analytics, ADLS Gen2, Delta Lake, Parquet, ETL/ELT Pipelines, Streaming Pipelines, Azure Event Hubs, Feature Engineering, Feast (Feature Store), Pandas, NumPy, Delta Lake, Medallion Architecture, Unity Catalog, Databricks Workflows
- **Cloud Platforms:** Microsoft Azure (incl. Azure OpenAI)
- **Databases:** PostgreSQL, MySQL, SQL Server, MongoDB
- **Monitoring & Security:** Prometheus, Grafana, Model Drift Detection, OpenTelemetry, Hallucination & Retrieval-Quality Monitoring, OAuth2.0, IAM, RBAC, Data Encryption, GDPR/HIPAA Compliance, Responsible AI Governance
- **BI & Visualization:** Power BI, Tableau, Matplotlib, Seaborn, Plotly
- **Version Control:** Git, GitHub, GitLab, Azure DevOps
- **Soft Skills:** Leadership, Communication, Problem Solving, Critical Thinking, Mentoring, Agile / Scrum Master

PROFESSIONAL EXPERIENCE

LeapGen.AI (Remote)

May 2026 - Present

Data and AI/ML Engineer

Project Overview: Multi-Model AI Chat Platform

Designed and contributed to a multi-tenant AI chat platform that allows users to log in and interact with multiple AI models (e.g., GPT, Claude) through a single unified interface, similar to ChatGPT. Built a centralized API gateway that routes requests to different LLM providers based on a model parameter, eliminating the need for separate integrations per model. Implemented a credit-based usage tracking system that maps token consumption to user credits, enabling metered billing and cost control.

Tech Stack: Azure Databricks, Azure Functions, Python, MongoDB, APIM, CosmosDB, Power BI, Python (FastAPI), OpenAI API, Anthropic API, unified via a single gateway service, Azure, Docker, CI/CD pipelines

Roles & Responsibilities:

- Designed the authentication flow supporting multiple OAuth providers (Gmail, Hotmail, etc.)
- Built and maintained a unified API gateway to abstract multiple LLM providers behind a single endpoint
- Implemented request routing logic based on the model name parameter, standardizing response formats across providers
- Developed a token-tracking and credit deduction system to support usage-based billing
- Built the chat dashboard UI, including chat history retrieval, session management, and model selection
- Collaborated with cross-functional and offshore teams using shared communication channels
- Participated in sprint planning, story prioritization, and timeline tracking with engineering leads

Project Overview: Seismic – Enterprise AI Healthcare & Telehealth Platform

Seismic is an enterprise AI-powered healthcare and telehealth platform that connects healthcare providers and patients through real-time video visits, digital patient onboarding, automated EHR synchronization, and an AI-driven clinical assistant. The platform is built on a microservices architecture spanning web, mobile, and backend layers, integrated with Azure cloud services and third-party healthcare systems (Athenahealth, Stream.io, Azure OpenAI). *(Project currently in progress.)*

Currently contributing to the platform's data and analytics layer, focused on building Databricks workflows for data processing/ML pipelines and developing Power BI dashboards for operational and clinical analytics.

Tech Stack: Python, SQL, FastAPI, Databricks, Microsoft Fabric, Power BI, Azure AI Search (RAG), Azure OpenAI, UpToDate (clinical grounding), ML pipelines (data-coffee-server, data-coffee-modelhub)

Role & Responsibilities

- Design and build Databricks workflows to process and transform clinical, operational, and usage data feeding the platform's analytics layer
- Develop data pipelines to ingest data from platform microservices (patient records, session/telehealth data, AI assistant interactions) into Databricks for downstream reporting
- Build and maintain Power BI dashboards for clinical and operational analytics, surfacing insights for stakeholders (e.g., patient engagement, telehealth session metrics, provider utilization)
- Collaborate with backend and ML teams (Python/FastAPI microservices, data-coffee ML pipelines) to ensure data models align with reporting requirements
- Work with Microsoft Fabric and Azure data services to optimize data flow between Databricks, Cosmos DB, and Power BI
- Partner cross-functionally with product and clinical stakeholders to define KPIs and translate them into dashboard views
- Support ongoing platform development by validating data accuracy and performance of analytics pipelines as new features roll out

Bank of America (Remote)

Jun 2025 – Jan 2026

Data and AI/ML Engineer Contract

Project: Enterprise Customer 360 Data Platform with ML-Driven Retention

Tech Stack: PySpark, Python, SQL, Azure Databricks, ADLS Gen2, Azure Data Factory, Delta Lake, Medallion Architecture, Unity Catalog, Databricks Feature Store, MLflow, Databricks Workflows

Roles & Responsibilities:

- Designed and built a Customer 360 lakehouse on Azure Databricks using a medallion (Bronze/Silver/Gold) architecture on Delta Lake, unifying customer data from core banking, cards, loans, investments, and digital channels.
- Built incremental ingestion with Auto Loader for batch files and Spark Structured Streaming from Azure Event Hubs, preserving raw data in Bronze for full audit and replay.
- Developed heavy PySpark transformations for standardization, deduplication, and PII masking in the Silver layer, enforcing column masks and row filters through Unity Catalog.
- Implemented identity resolution (blocking, fuzzy matching, graph-based grouping) to assign a stable master customer ID and produce a single golden record using survivorship rules.
- Built curated gold feature tables and business attributes, and registered features in the Databricks Feature Store to eliminate train/serve skew.

- Delivered a churn-prediction model (XGBoost) with MLflow tracking and Model Registry, tuned on AUC-PR for imbalanced data, and deployed batch scoring plus an optional real-time Model Serving endpoint.
- Added Lakehouse Monitoring for data and prediction drift with automated retraining triggers, and orchestrated end-to-end pipelines with Databricks Workflows.
- Enforced governance and compliance (RBAC, PII masking, full lineage and audit) to meet KYC/AML and model-risk expectations, and pushed the unified view back to CRM and BI for marketing and call-center use.

TD Bank, Ewing Township, NJ (Remote)

Oct 2023 – Apr 2025

Machine Learning / AI Engineer Contract

Project: AI-Powered KYC Document Intelligence & Verification System

Tech Stack: Azure Databricks, PySpark, Python, TensorFlow, FastAPI, Docker, MLflow, Azure, LangChain, ChromaDB/Pinecone, PostgreSQL

Roles & Responsibilities:

- Built an end-to-end production system for automated KYC document processing, using computer vision and Generative AI to classify, validate, and extract insights from passports, driver's licenses, and financial records.
- Architected a FastAPI + Docker REST API around a fine-tuned MobileNetV2 backbone, enabling real-time document classification with sub-200ms latency for customer-facing banking applications.
- Designed a RAG pipeline via LangChain that feeds classified outputs and extracted entities into an LLM to generate fraud-risk scores, transaction explanations, and compliance summaries.
- Integrated MLflow experiment tracking and a centralized Model Registry to manage Staging→Production transitions, ensuring auditability and regulatory compliance.
- Built scalable PySpark pipelines on Azure Databricks to ingest, clean, validate, and transform high-volume document metadata for model training and RAG retrieval.
- Developed an automated data-validation framework for noisy inputs (blurred scans, skewed images, OCR inconsistencies), reducing production inference errors and manual intervention.
- Engineered a CI/CD pipeline (GitHub Actions, Azure DevOps) automating unit and integration testing, enabling zero-downtime Blue-Green deployments in a highly regulated environment.
- Fine-tuned small language models with LoRA and adapter techniques for low-latency, cost-efficient financial insights while preserving data quality.
- Implemented Prometheus/Grafana and OpenTelemetry observability to track drift, latency, token usage, and retrieval accuracy, enabling proactive retraining and rapid incident response.
- Optimized inference cost and throughput via model quantization, request batching, caching and latency-throughput tuning, plus an async Celery/Redis pipeline for heavy OCR workloads.
- Enforced secure access with IAM, RBAC, OAuth2 and secrets management, and operationalized LLM evaluation and governance (benchmarking, hallucination detection, bias analysis, human-in-the-loop validation).

Evolent, Arlington, VA (Remote)

Nov 2021 – Jun 2023

Data Scientist / ML Engineer Contract

Project: Enterprise Phishing Intelligence & Automated Risk Mitigation

Tech Stack: Python (Scikit-learn, XGBoost), Apache Spark / PySpark, SQL Server, MLflow, Airflow, SHAP, Docker, Tableau

Roles & Responsibilities:

- Engineered a predictive threat-detection ecosystem, shifting security operations from reactive pattern-matching to proactive ML-driven defense with a strictly controlled false-positive rate.
- Implemented a cascading inference architecture — a lightweight heuristic pre-filter routing complex edge cases to a high-fidelity XGBoost ensemble — to optimize high-volume traffic.
- Built an Airflow-orchestrated retraining pipeline with human-in-the-loop analyst labels to counter concept drift and evolving zero-day tactics.
- Developed lexical and behavioral features (URL Shannon entropy, domain-age cross-referencing, n-gram TF-IDF) to detect homograph and obfuscation attacks.
- Integrated SHAP for transparent, feature-level risk indicators, reducing analysts' mean time to remediation.
- Architected Apache Spark ETL over multi-terabyte communication logs with a centralized feature store, eliminating training-serving skew.
- Calibrated model probabilities (Platt scaling, isotonic regression) for reliable automated blocking policies, and hardened the pipeline with adversarial testing and synthetic augmentation.
- Established MLflow-based monitoring for calibration and feature drift, and authored MLOps playbooks for rollbacks, canary A/B deployments, and incident response.
- Fine-tuned BERT/DistilBERT with HuggingFace Transformers for escalation detection, and built spaCy/NLTK NLP pipelines for keyword extraction and clustering across communication logs.

FEATURED PROJECTS

AI Resume Analyzer & Interview Preparation Platform

Tech Stack: Python, Streamlit, Google Gemini 2.5 Flash, LangChain, FAISS, HuggingFace Embeddings, PyPDF

- Built an end-to-end GenAI platform combining RAG, semantic search, and LLMs to analyze resumes and coach candidates for interviews in real time.
- Delivered ATS resume scoring, resume- and job-description-grounded technical and HR question generation, and a conversational AI interviewer with real-time answer evaluation.
- Architected a modular RAG pipeline: PyPDF extraction, recursive chunking, HuggingFace embeddings indexed in FAISS, and LangChain-orchestrated Gemini 2.5 Flash prompting, with each capability shipped as an isolated module behind a Streamlit UI.

Resident Complaint Auto-Routing System

Tech Stack: Python, TF-IDF, Cosine Similarity, MongoDB, Streamlit

- Developed an NLP system that reads free-text resident complaints, predicts the correct team, and routes them in seconds.
- Added confidence scoring with a human-in-the-loop safety net for low-confidence cases, plus end-to-end request tracking and resident outcome ratings; live demo deployed on Streamlit.

EDUCATION

Master of Science in Computing and Data Science

Advanced coursework in machine learning, data engineering and applied AI.

Master of Business Administration

Business strategy, operations and analytics-driven decision-making.

Bachelor of Commerce

Foundation in finance, accounting and quantitative methods.

CERTIFICATIONS & CONTINUOUS LEARNING

- Scrum Master Certified (SMC).
- CompTIA IT Fundamentals (ITF+).
- DataCamp: 100,048 XP across 24 completed courses, including Large Language Models (LLMs), Implementing AI Solutions in Business, Understanding Machine Learning, Data Manipulation with pandas, Intermediate SQL, Database Design, Snowflake SQL, Data Warehousing, Cloud Computing and Understanding Data Engineering.

ACHIEVEMENTS

- Delivered 15+ production ML, GenAI and data-engineering projects across finance, security, healthcare, real estate and analytics.
- Solved 200+ LeetCode problems in Python and SQL, with strong focus areas in SQL queries & joins (90%) and arrays & hashing (85%).
- Accumulated 100,048 DataCamp XP across 24 courses, demonstrating sustained, self-driven upskilling.
- Designed and deployed a full production portfolio platform (Next.js, Tailwind, Framer Motion) featuring live projects and a live GitHub feed.
- Mentors aspiring data professionals through one-on-one tutoring in Python, SQL, ML and interview preparation.